

Problem 101. Consider the sequence of rational numbers defined by

$$x_1 = \frac{4}{3} \quad \text{and} \quad x_{n+1} = \frac{x_n^2}{x_n^2 - x_n + 1}, \quad \text{for } n \geq 1.$$

If the rational number $\sum_{k=1}^n x_k$ is expressed in lowest terms for $n \geq 1$, show that its numerator is a perfect square.

Problem 102. Let BM be a median in an acute-angled triangle ABC . A point K is chosen on the line through C tangent to the circumcircle of $\triangle BMC$ so that $\angle KBC = 90^\circ$. The segments AK and BM meet at J .

Prove that the circumcentre of $\triangle BJK$ lies on the line AC .

Problem 103. Let ABC be an acute-angled triangle. The line through C perpendicular to AC meets the external angle bisector of $\angle ABC$ at D . Let H be the foot of the perpendicular from D onto BC . The point K is chosen on AB so that $KH \parallel AC$. Let M be the midpoint of AK .

Prove that $MC = MB + BH$.

Problem 104. The plane is divided into unit squares by means of two sets of parallel lines. The unit squares are coloured in 1201 colours so that no rectangle of perimeter 100 contains two squares of the same colour.

Show that no rectangle of size 1×1201 contains two squares of the same colour.